

Empirical Proof Record: VALIDATED

2026-05-20T15:34:41.416760+00:00

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Proof ID: 5c5f94eb3df2a64297e1d51b2e085d6a...

Created: 2026-05-20T15:34:41.416760+00:00

1. Claim

PyMC and NumPyro fit the same NormalMean model on the same synthetic Normal($\mu=0.5$, $\sigma=0.2$) data ($N=200$, $\text{seed}=20260517$) and produce posteriors that agree within tolerance: $|\mu_{\text{pymc}} - \mu_{\text{numpyro}}| \leq 0.1$ AND $|\text{width_ratio} - 1| \leq 0.5$. (RFC 0002 Phase E1: cross-framework validation.)

- **Operationalisation:** Generate N samples from $\text{Normal}(\text{true_mu}, \text{true_sigma})$ with fixed seed. Fit the same model ($\mu \sim \text{Normal}(0, 10)$; $\sigma \sim \text{HalfNormal}(1)$; $y \sim \text{Normal}(\mu, \sigma)$) under both backends. Extract $\text{posterior_mean}(\mu) + 94\% \text{ HDI}(\mu)$. Compute $\text{mean_diff} = |\mu_{\text{pymc}} - \mu_{\text{numpyro}}|$ and $\text{width_ratio} = \text{hdi_width_pymc} / \text{hdi_width_numpyro}$. VALIDATED iff both tolerances are satisfied (binary; 1.0 / 0.0). RE-FUTED if either bound is breached.
- **Threshold:** `cross_framework_agreement >= 1.0 proportion`
- **H0:** PyMC and NumPyro disagree on posterior mean and/or HDI width beyond the documented sampler-noise tolerance — at least one backend is broken or the model is mis-specified.
- **H1:** Both backends produce statistically equivalent posteriors on the same data + model — corroborates the framework's Bayesian inference machinery.

2. Verdict

VALIDATED — observed 1.0

PyMC $\mu=0.4995$, $\text{HDI}=[0.4696, 0.5261]$; NumPyro $\mu=0.4993$, $\text{HDI}=[0.4709, 0.5277]$. $\text{mean_diff}=0.0003$ (≤ 0.1 tol); $\text{width_ratio}=0.9965$ (within ± 0.5).

3. Evidence

Pillar	Statistic	Value	95% CI	Library
bayesian_cross_framework_pymc_numpyro	cross_framework_agreement	1.0	—	pymc 6.

4. Pre-registration

- Registered at: 2026-05-20T15:34:41.167155+00:00
- Config hash: 201b9d70c5b338371d95356a52bbdaab...
- Data hash: 201b9d70c5b338371d95356a52bbdaab...

5. Signature

0602513105b54f6e0c5d0823f5dad52ba7e8620d18a808c67f40d6a547c51815